Experiment 3

AIM: Develop Expense Tracking Module with Exception Handling.

Source: Program.cs

Code:

```
using System;
using System.Collections.Generic;

namespace Practical3
{
    class Expense
    {
        public int ExpenseId;
        public string Category;
        public double Amount;
        public string PaymentMode;
        public DateTime ExpenseDate;
        public void AddExpense()
        {
            Console.Write("Enter Expense ID: ");
            ExpenseId = Convert.ToInt32(Console.ReadLine());

            Console.Write("Enter Category: ");
            Category = Console.ReadLine();

            Console.Write("Enter Amount: ");
            Amount = Convert.ToDouble(Console.ReadLine());

            if (Amount <= 0)
            {
                throw new Exception("Amount must be greater than 0.");
            }

            Console.Write("Enter Payment Mode: ");
            PaymentMode = Console.ReadLine();

            Console.Write("Enter Expense Date (dd/mm/yyyy): ");
            ExpenseDate = Convert.ToDateTime(Console.ReadLine());


        }
        public void DisplayExpense()
        {
            Console.WriteLine("\n----- Expense Details -----");
            Console.WriteLine("Expense ID   : " + ExpenseId);
            Console.WriteLine("Category     : " + Category);
            Console.WriteLine("Amount       : " + Amount);
            Console.WriteLine("Payment Mode : " + PaymentMode);
            Console.WriteLine("Expense Date : " +
ExpenseDate.ToShortDateString());
        }
```

```
}
class Program
{
    static void Main(string[] args)
    {
        List<Expense> expenses = new List<Expense>();
        int choice;

        do
        {
            Console.WriteLine("\n===== Expense Tracking System =====");
            Console.WriteLine("1. Add Expense");
            Console.WriteLine("2. View All Expenses");
            Console.WriteLine("3. Exit");

            Console.Write("Enter Your Choice: ");
            choice = Convert.ToInt32(Console.ReadLine());

            try
            {
                switch (choice)
                {
                    case 1:
                        Expense e = new Expense();
                        e.AddExpense();
                        expenses.Add(e);
                        Console.WriteLine("Expense Added Successfully.");
                        break;

                    case 2:
                        if (expenses.Count == 0)
                        {
                            Console.WriteLine("No Expense Found.");
                        }
                        else
                        {
                            foreach (Expense exp in expenses)
                            {
                                exp.DisplayExpense();
                            }
                        }
                        break;

                    case 3:
                        Console.WriteLine("Thank You!");
                        break;

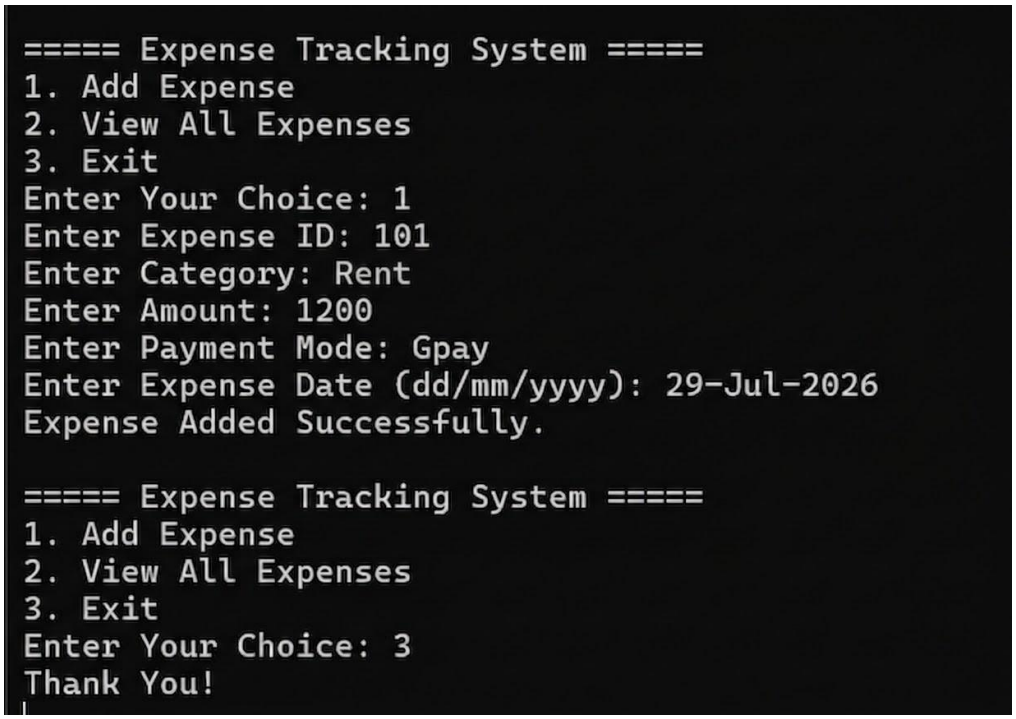
                    default:
                        Console.WriteLine("Invalid Choice.");
                        break;
                }
            }
            catch (Exception ex)
            {
                Console.WriteLine("Error: " + ex.Message);
```

```
                }

            } while (choice != 3);

            Console.ReadKey();
        }
    }
}
```

Output:-



GitHub: [Link]